INCH-POUND

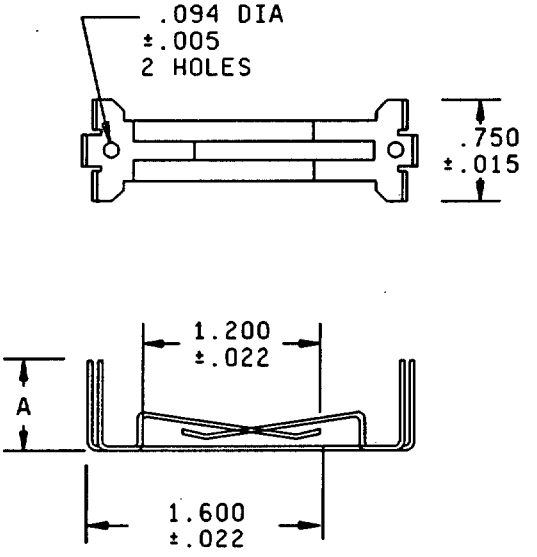
MIL-H-87111/7A
28 March 1994
SUPERSEDING
MIL-H-87111/7
7 September 1984

MILITARY SPECIFICATION SHEET

HEAT SINKS, SEMICONDUCTOR DEVICES, CLASS 1, (FOR DUAL-IN-LINE PACKAGES)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-H-87111.



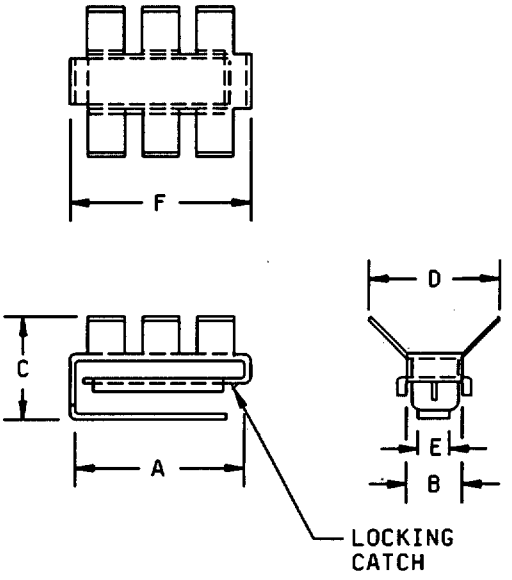
Dash no. M87111/07- (see note 3)	No. of pins in package	Dimension A
1*01	20-40	.250 (6.35)
1*02	20-40	.500 (12.70)
1*03	20-40	.750 (19.05)

CONFIGURATION A

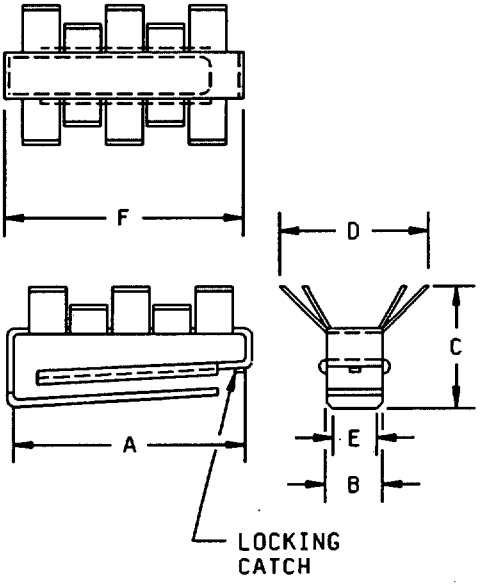
FIGURE 1. Heat sinks (dual-in-line packages).

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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CONFIGURATION B



CONFIGURATION C

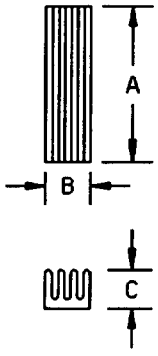
FIGURE 1. Heat sinks (dual-in-line packages) - Continued.

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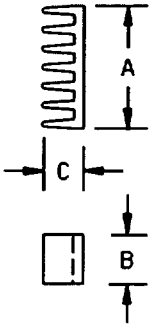
Dash number M87111/07- (see notes 3 and 4)	No. of pins in package	Config-uration	Dimensions						Number of fins
			A	B	C	D	E	F	
1*04	14/16	B	.837 (21.26)	.250 (6.35)	.410 (10.41)	.600 (15.24)	.200 (5.08)	.887 (22.53)	6
1*23	14/16	B	.837 (21.26)	.310 (7.87)	.410 (10.41)	.600 (15.24)	.200 (5.08)	.887 (22.53)	6
1*05	18	C	.930 (23.62)	.250 (6.35)	.556 (14.12)	.700 (17.78)	.210 (5.33)	1.125 (28.58)	10
1*06	20	C	1.055 (26.80)	.250 (6.35)	.556 (14.12)	.700 (17.78)	.210 (5.33)	1.125 (28.58)	10
1*07	24	C	1.280 (32.51)	.540 (13.72)	.580 (14.73)	1.008 (25.60)	.480 (12.19)	1.460 (37.03)	12
1*08	40	C	2.090 (53.09)	.525 (13.34)	.590 (14.99)	.980 (24.89)	.480 (12.19)	2.256 (57.30)	20

FIGURE 1. Heat sinks (dual-in-line packages) - Continued.

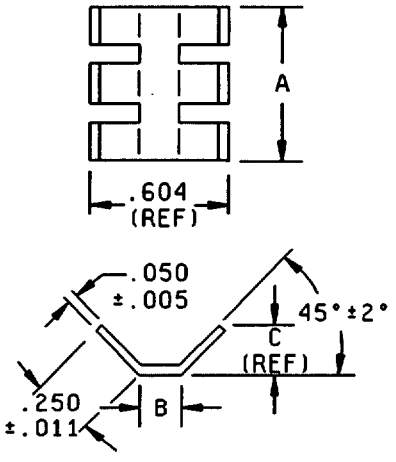
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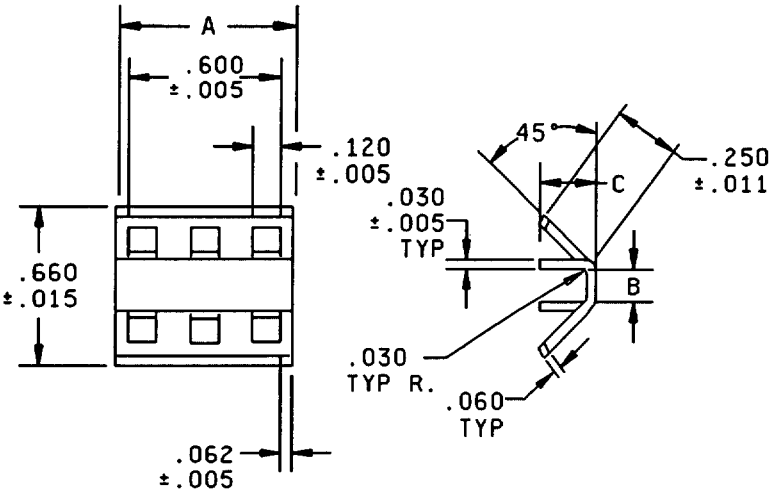
CONFIGURATION D



CONFIGURATION E



CONFIGURATION F



CONFIGURATION G

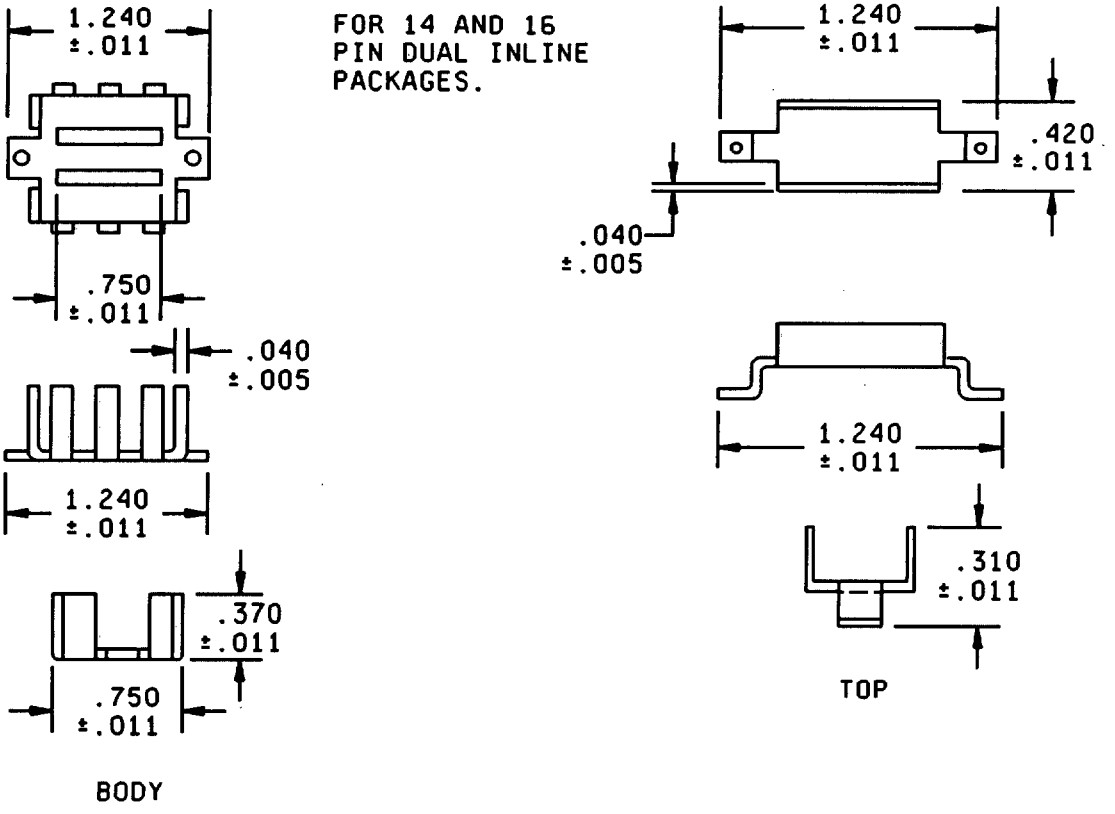
FIGURE 1. Heat sinks (dual-in-line packages) - Continued.

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Dash number M87111/07-	No. of pins in package	Configuration (see note 5)	Dimensions			Number of fins
			A (see note 7)	B ±.011 (0.28)	C ±.011 (0.28)	
1A09	14/16	D	.375 (9.53)	.250 (6.35)	.200 (5.08)	4
1A10	14/16	D	.750 (19.05)	.210 (5.33)	.240 (6.10)	3
1A11	14/16	D	.750 (19.05)	.250 (6.35)	.190 (4.83)	4
1A12	14/16	D	.750 (19.05)	.400 (10.16)	.240 (6.10)	5
1A13	24	D	1.250 (31.75)	.530 (13.46)	.190 (4.83)	5
1A14	28	D	1.450 (36.83)	.530 (13.46)	.190 (4.83)	5
1A15	40	D	2.000 (50.80)	.530 (13.46)	.190 (4.83)	5
1A16	64	D	3.200 (81.28)	.530 (13.46)	.190 (4.83)	5
1A17	14/16	E	.400 (10.16)	.250 (6.35)	.240 (6.10)	5
1A18	14/16	E	.730 (18.54)	.250 (6.35)	.190 (4.83)	7
1A19	14/16	E	.750 (19.05)	.250 (6.35)	.240 (6.10)	9
1A20	14/16	E	.750 (19.05)	.400 (10.16)	.240 (6.10)	9
1*21 (see note 3)	14/16	F (see note 6)	.750 (19.05)	.250 (6.35)	.212 (5.38)	NA
1A22	14/16	G	.750 (19.05)	.250 (6.35)	.230 (5.84)	NA
1A23	14/16	D	.750 (19.05)	.495 (12.57)	.240 (6.10)	5

FIGURE 1. Heat sinks (dual-in-line packages) - Continued.

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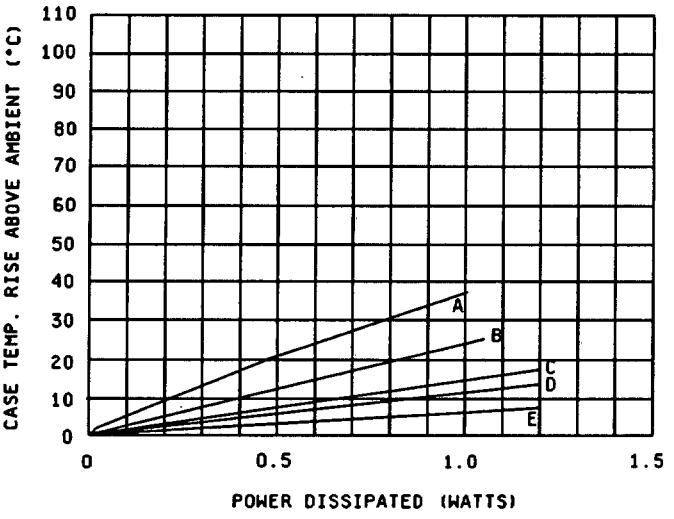
CONFIGURATION H
(SEE NOTE 8)

M87111/07-1H23

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses and are given for general information only.
3. An asterisk in the dash number represents finish. See finish and PIN paragraphs herein.
4. Configurations B and C are available in black anodized and gold anodized finishes only.
5. Configurations A, D, E, and G are available in black anodize finish only.
6. Configuration F is available in black anodize, gold chromate, and pre-black anodize finishes only.
7. Configurations D, E, F, and G only: For lengths (dimension A) up to 1.000 (25.40 mm), tolerance is ±.015 (0.38 mm). For lengths over 1.000, tolerance is ±.022 (0.56 mm).
8. Configuration H is available in hard anodize finish only.
9. All configurations: Unless otherwise indicated, tolerance is ±.017 (0.43 mm).

FIGURE 1. Heat sinks (dual-in-line packages) - Continued.

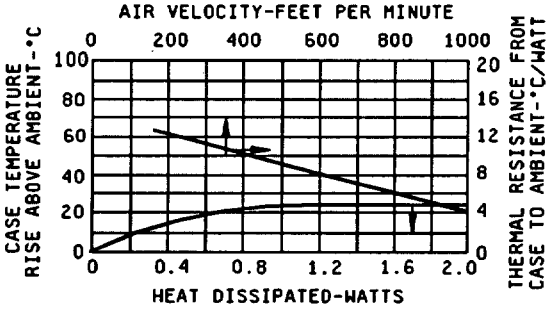
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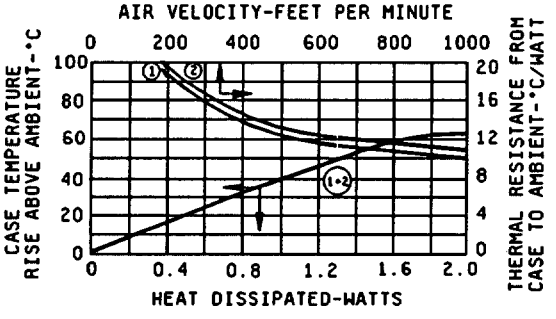
DESCRIPTION OF CURVES

A. N.C. HORIZ. DEVICE ONLY MOUNTED TO G-10.
B. N.C. HORIZ AND VERT. WITH DISSIPATOR.
C. 200 FPM WITH DISSIPATOR.
D. 500 FPM WITH DISSIPATOR.
E. 1000 FPM WITH DISSIPATOR.

CONFIGURATION A



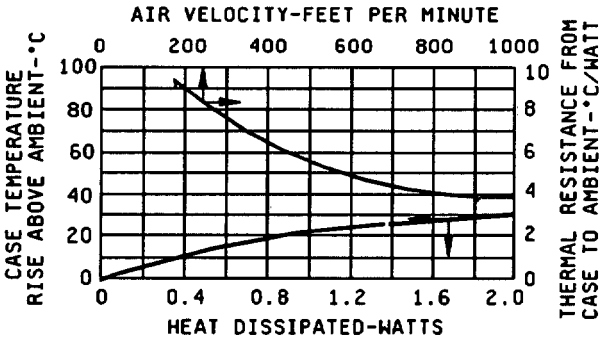
CONFIGURATION B



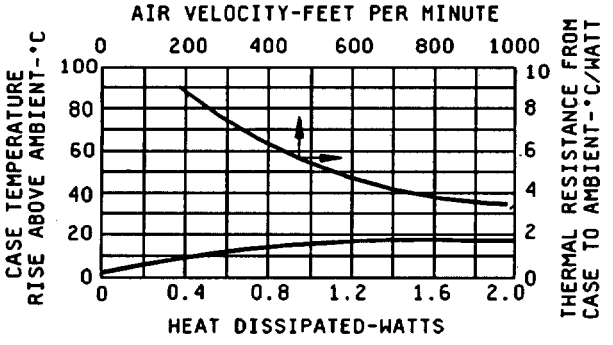
CONFIGURATION C
M87111/07-1✱05
M87111/07-1✱06

FIGURE 2. Thermal characteristics.

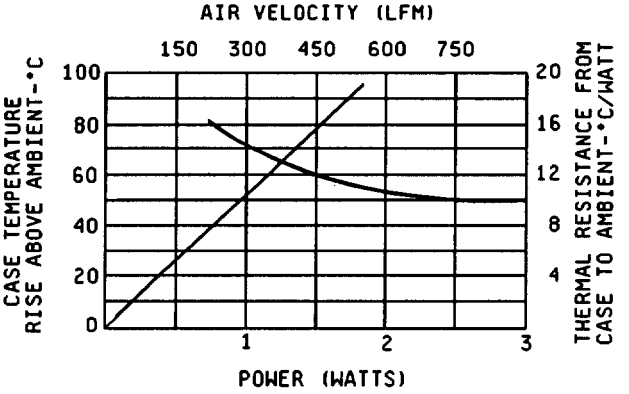
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CONFIGURATION C
M87111/07-1*07



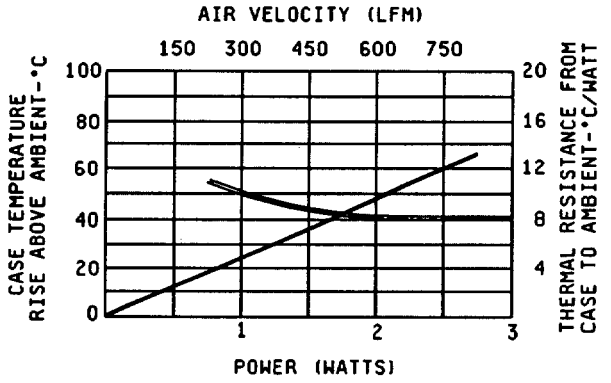
CONFIGURATION C
M87111/07-1*08



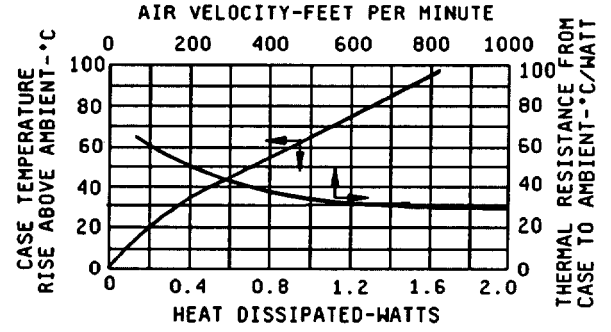
CONFIGURATION D
M87111/07-1A09

FIGURE 2. Thermal characteristics - Continued.

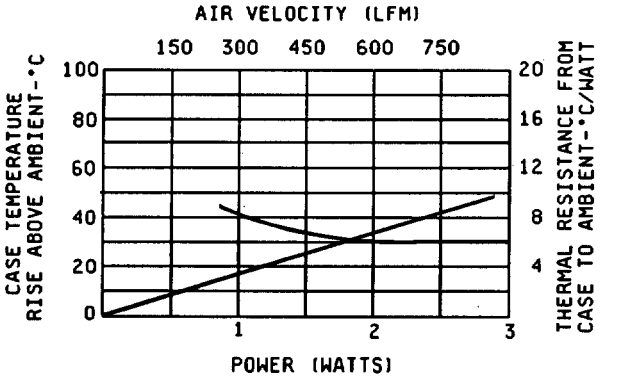
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CONFIGURATION D
M87111/07-1A10



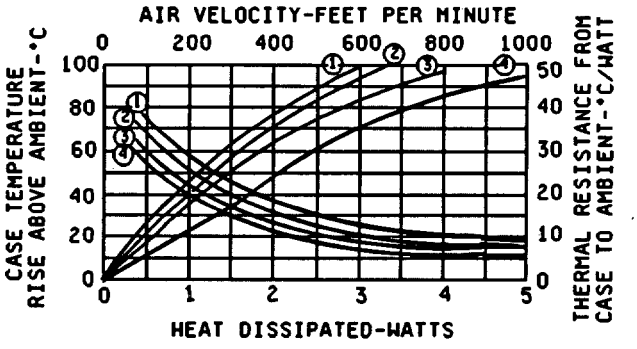
CONFIGURATION D
M87111/07-1A11



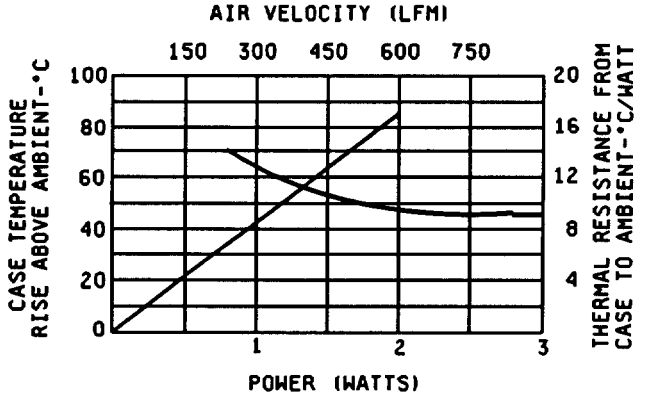
CONFIGURATION D
M87111/07-1A12

FIGURE 2. Thermal characteristics - Continued.

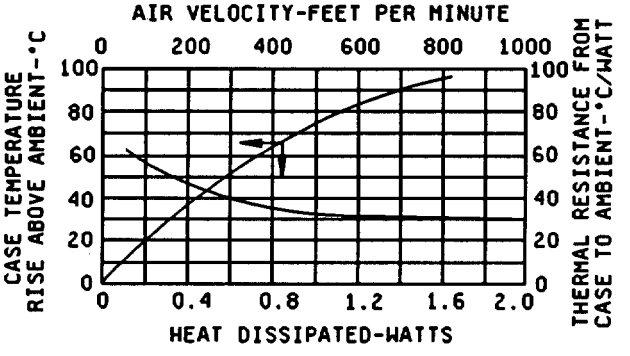
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CONFIGURATION D
M87111/07-1A13 M87111/07-1A15
M87111/07-1A14 M87111/07-1A16



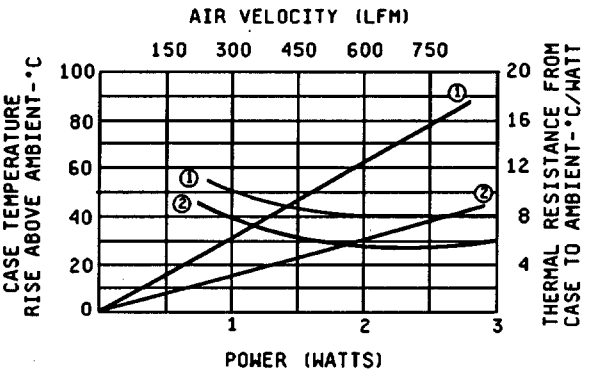
CONFIGURATION E
M87111/07-1A17



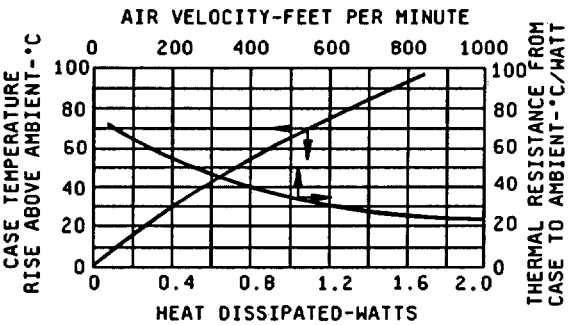
CONFIGURATION E
M87111/07-1A18

FIGURE 2. Thermal characteristics - Continued.

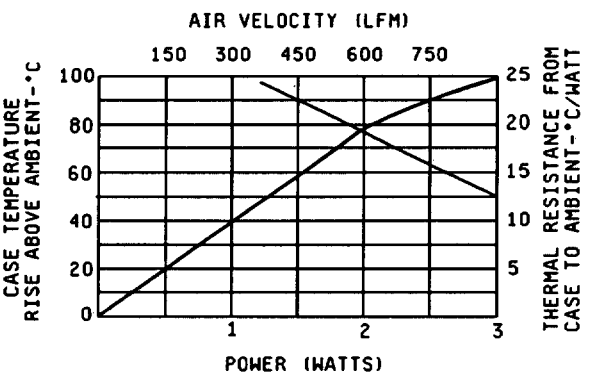
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CONFIGURATION E
M87111/07-1A19
M87111/07-1A20



CONFIGURATION F
M87111/07-1*21



CONFIGURATION G
M87111/07-1A22

FIGURE 2. Thermal characteristics - Continued.

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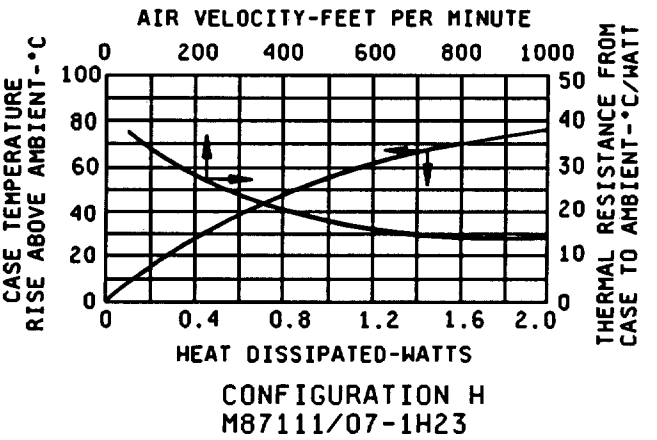


FIGURE 2. Thermal characteristics - Continued.

REQUIREMENTS:

Dimensions and configurations: See figure 1.

Material: Aluminum alloy (class 1).

Finish options:

Black anodize (A).

Hard anodize (H).

Gold anodize (L) conforming to type I or II of MIL-A-8625.

Pre-black anodize (P) conforming to type II, class II, of MIL-A-8625 (pre-anodized coil stock).

Gold chromate (R).

Thermal resistance: See figure 2.

Mounting surface roughness: Not applicable.

Surface roughness: Not applicable.

JEDEC case size: Not applicable.

Number of fins: See figure 1.

Part or Identifying Number (PIN): M87111/07- (plus dash number from figure 1). The asterisk in the PIN represents the finish letter (see above). PIN example: M87111/07-1A01.

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Supersession data: The original edition of this specification sheet incorrectly listed the letter B in the dash number tables and the finish paragraph to indicate black anodize finish. Revision A changes the letter B to an A to correctly identify black anodize finish as shown in the basic specification (MIL-H-87111). Finish G (gold plating in accordance with MIL-H-87111) has been deleted from this specification sheet and replaced by new finish L (gold anodize).

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - AS, CG, MC, SH, TD
Air Force - 19, 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5999-0311)

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